

The $\mathfrak{B}_2 \times \mathfrak{C}_1$ row: check, and where the UV is

Everything verified against the quoted vertices; one substantive error; three logarithms, only two of them real

$$\eta = \frac{p^+}{k^+}, \quad \bar{\eta} = 1 - \eta, \quad q = k^+ - p^+, \quad u = z - x, \quad R = y - x, \quad V = \frac{x' - w'}{(x' - w')^2}, \quad C_{UV}(\eta) = \eta \bar{\eta} + \frac{\eta}{\bar{\eta}} + \frac{\bar{\eta}}{\eta} = \frac{\mathcal{P}_{gg}(\eta)}{2N_c}$$

Verdict. The algebra of eq. (2) is **exactly right**: all six tensor blocks, the $d_\perp$, the prefactor $\frac{g^4}{8\pi^4} \frac{p^+(k^+ - p^+)}{k^{+3}}$, the $\delta^{(2)}$, and the Λ cutoff. The w -integration of eq. (3) is right too, and all six p^+ weights transfer correctly. **One substantive error:** the colour collapse $f^{adc} f^{ebf} U^{db}(x) U^{ce}(z) \rightarrow -N_c U^{af}(x)$ is true only when $U(z) = U(x)$ — it silently drops the z -dependence of the Wilson line, and that is not an identity. **The UV is at $z \rightarrow x$** , logarithmic, with residue exactly $C_{UV}(\eta) = \frac{\mathcal{P}_{gg}(\eta)}{2N_c}$ times the product of the two LO Weizsäcker–Williams kernels — and all the $d_\perp$'s cancel out of it.

1. The tensor bracket of eq. (2) is exact

Building $\mathfrak{B}_2$ from its definition with the substitutions that eq. (1) calls for — $(\mathbf{x}, \mathbf{z}, \mathbf{w}, k^+, i)_{\text{def}} \rightarrow (y, x, z, k^+ - p^+, k)$, so that the definition's $(p^+ + k^+)$ becomes $p^+ + (k^+ - p^+) = k^+$ — gives B_{jk} . Contracting with $\mathfrak{C}_{1jki}$'s tensor,

$$\mathcal{E}_{im} = \delta_{im} \text{tr } B - \frac{k^+}{p^+} B_{mi} - \frac{k^+}{k^+ - p^+} B_{im},$$

and comparing block by block against the six terms you wrote:

$d_\perp$	max derived	max quoted	max difference
2	3.47×10^2	3.47×10^2	2.8×10^{-14}
3	1.01×10^2	1.01×10^2	2.8×10^{-14}
4	7.27×10^1	7.27×10^1	2.1×10^{-14}
5	5.72×10^1	5.72×10^1	2.1×10^{-14}

300 random configurations each. Two details worth naming, because they are easy to get wrong and you got them right:

- The $d_\perp$ in the first block is δ^{jj} from $\text{tr } B$ — it belongs only there.
- The third pieces of your $\frac{k^+ \delta_{im}}{p^+}[\dots]$ and $\frac{k^+ \delta_{im}}{k^+ - p^+}[\dots]$ blocks (the $-\frac{(y-x)^2 - (y-z)^2}{2(x-z)^2}$ terms) are *not* part of $\text{tr } B$. They come from the δ^{jk} piece of B inside $-\frac{k^+}{p^+} B_{mi}$ and $-\frac{k^+}{q} B_{im}$. Grouping them the way you did is correct.

2. The prefactor

$$\mathfrak{A}^\dagger = \frac{-ig}{\sqrt{2}\pi\sqrt{k^+}}, \quad \mathfrak{C}_1^\dagger = \frac{-g}{2\pi\sqrt{2}k^+} \frac{\sqrt{p^+q}}{k^+}, \quad \mathfrak{B}_2 = \frac{ig^2\sqrt{p^+q}}{4\pi^2k^+} \frac{1}{p^+(y-x)^2 + q(y-z)^2}$$

$$\Rightarrow \frac{2}{(2\pi)^3} \times (\text{product}) = -\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{p^+(k^+ - p^+)}{k^{+3}} \cdot \frac{1}{[\dots]}$$

Magnitude, k^+ power and $p^+(k^+ - p^+)$ all match eq. (2) exactly. The sign comes out -1 ; it is restored by the first bracket, since $-\bar{\mathfrak{A}}^{\dagger a} + \mathfrak{A}^{\dagger b'} U^{b'a}(w') \propto -[U^{ab'}(x') - U(w')]$. Two caveats there: $\bar{\mathfrak{A}}$, $\mathfrak{B}$, $\bar{\mathfrak{C}}$ are used but never defined in what you sent, so I assumed $\bar{\mathfrak{A}}^a \supset U^{ab}(x)\rho^b(x)$; and that construction gives $U^{b'a}(w')$, while eq. (2) writes $U^{ab'}(w')$ — one of the two needs transposing.

3. The w -integration of eq. (3)

$\mathfrak{C}_1$'s $\delta^{(2)}(z_{\text{arg}} + \frac{p^+}{k^+} x_{\text{arg}})$ with $z_{\text{arg}} = w - z$, $x_{\text{arg}} = z - x$ is exactly your $\delta^{(2)}[w - z + \frac{p^+}{k^+}(z - x)]$, and it sets $w = z - \frac{p^+}{k^+}(z - x)$, so

$$-ik \cdot (w' - w) = -ik \cdot (w' - z) - i \frac{p^+}{k^+} k \cdot (z - x),$$

the two phases of eq. (3). Correct. The six p^+ weights also transfer exactly (eq. (3)'s bracket is eq. (2)'s times $\frac{p^+(k^+ - p^+)}{k^{+2}}$, with $\frac{1}{k^+}$ pulled out front):

block	eq. (2) coeff $\times \frac{p^+q}{k^{+3}}$	eq. (3) coeff $\times \frac{1}{k^+}$
T1	$\frac{p^+q}{k^{+3}}$	$\frac{p^+q}{k^{+3}}$
T2	$-\frac{p^+}{k^+q}$	$-\frac{p^+}{k^+q}$
T3	$\frac{q}{k^{+2}}$	$\frac{q}{k^{+2}}$
T4	$-\frac{q}{k^+p^+}$	$-\frac{q}{k^+p^+}$
T5	$\frac{p^+}{k^{+2}}$	$\frac{p^+}{k^{+2}}$
T6	$-\frac{1}{k^+}$	$-\frac{1}{k^+}$

But eq. (3) still writes $\int_{w,w'}$. The $\delta^{(2)}$ ate the w integral — it should be $\int_{w'}$.

4. The colour algebra: one exact, one not

Using genuine adjoint $SU(3)$ Wilson lines $U^{ab} = 2 \text{tr}(t^a V t^b V^\dagger)$ (a generic $SO(8)$ matrix will *not* do — the identity $f^{def} U^{da} U^{eb} U^{fc} = f^{abc}$ fails for it):

Exact: $f^{abc} f^{cbd} = -N_c \delta^{ad}$ (checked to 10^{-15}), so $-f^{abc} f^{cbd} U^{df}(y) = +N_c U^{af}(y)$. Your second colour term is right. (The δ^{ad} you kept is redundant — just write $N_c U^{af}(y)$.)

Not exact: $f^{adc} f^{ebf} U^{db}(x) U^{ce}(z)$.

	$\ M + N_c U^{af}(x)\ $	$\ M + N_c U^{fa}(x)\ $
$U(z) = U(x)$	1.3×10^{-15}	5.1
$U(z)$ free	2.8	2.8

(for scale, $\|M\| = 1.35$, $N_c \|U\| = 2.58$.) So the collapse $\rightarrow -N_c U^{af}(x)$ holds **only** at $U(z) = U(x)$. Your index order U^{af} is the right one — the transpose fails — but the step itself is an approximation, not an identity.

This matters twice over. It is **exactly right in the UV region** $z \rightarrow x$, which is what §5 is about — so if the goal is to extract the UV pole, the collapse is legitimate and you should say so. It is **wrong everywhere else**, and in particular it destroys the large- $|z|$ behaviour, where $U(z) \rightarrow 1$ rather than $U(x)$, and where the cancellation against the $\mathfrak{B}_2$ term has to happen (§6). I would keep $U(z)$ in eq. (3) and take $U(z) \rightarrow U(x)$ only inside the UV subtraction.

5. The UV term: $z \rightarrow x$

Put $u = z - x \rightarrow 0$ at fixed y , and remember $(x - z) = -u$ (that sign is where hand calculations go wrong). Only four of the six blocks are singular, and each contributes after contracting with $V^i \frac{u^m}{u^2}$ and averaging over $\hat{u}$, using $\langle u^m u^n \rangle = \frac{u^2}{d_\perp} \delta^{mn}$:

block	singular part	after contraction and $\hat{u}$ -average
T1	$+d_\perp \delta_{im} \frac{R \cdot u}{u^2}$	$+1 \cdot \frac{V \cdot R}{u^2}$
T2	$+\frac{k^{+2}}{q^2} \frac{R^i u^m}{u^2}$	$+\frac{k^{+2}}{q^2} \frac{V \cdot R}{u^2}$
T3	$-\frac{2k^+}{p^+} \delta_{im} \frac{R \cdot u}{u^2}$	$-\frac{2k^+}{d_\perp p^+} \frac{V \cdot R}{u^2}$
T4	$+\frac{k^{+2}}{p^{+2}} \frac{R^i u^m}{u^2}$	$+\frac{k^{+2}}{p^{+2}} \frac{V \cdot R}{u^2}$
T5	$-\frac{2k^+}{q} \delta_{im} \frac{R \cdot u}{u^2}$	$-\frac{2k^+}{d_\perp q} \frac{V \cdot R}{u^2}$
T6	$+\frac{2k^{+2}}{p^+ q} \frac{R^m u^i}{u^2}$	$+\frac{2k^{+2}}{d_\perp p^+ q} \frac{V \cdot R}{u^2}$

The four $\frac{1}{d_\perp}$ pieces cancel identically, because

$$\frac{k^{+2}}{p^+ q} - \frac{k^+}{p^+} - \frac{k^+}{q} = \frac{k^{+2} - k^+(q + p^+)}{p^+ q} = 0.$$

What survives is $1 + \frac{k^{+2}}{q^2} + \frac{k^{+2}}{p^{+2}}$, and the $\frac{p^+ q}{k^{+2}}$ weight turns that into $\eta \bar{\eta} + \frac{\eta}{\bar{\eta}} + \frac{\bar{\eta}}{\eta}$:

$$\text{integrand} \xrightarrow{z \rightarrow x} \frac{1}{k^+} C_{\text{UV}}(\eta) \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} \frac{1}{(z - x)^2}, \quad C_{\text{UV}}(\eta) = \frac{\mathcal{P}_{gg}(\eta)}{2N_c}$$

Numerically, $u^2 \times \langle \text{integrand} \rangle_{\hat{u}}$ against the prediction:

$d_\perp$	η	$ u = 10^{-2}$	10^{-3}	10^{-4}	predicted
2	0.30	0.828730	0.829104	0.830506	0.829464
2	0.65	0.974290	0.972616	0.972966	0.972507
3	0.30	0.076316	0.075325	0.075430	0.075753
3	0.65	-0.461266	-0.461669	-0.460500	-0.460978
4	0.30	-0.140156	-0.138162	-0.139394	-0.139054
4	0.65	-0.156533	-0.156899	-0.156639	-0.156479

With $d^2 z = |u| d|u| d\phi$ this is $\int \frac{d|u|}{|u|}$: **a single logarithmic UV divergence**. So the divergent part of the row is

$$\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \int_{\Lambda}^{k^+ - \Lambda} \frac{dp^+}{2\pi} \frac{\mathcal{P}_{gg}(\frac{p^+}{k^+})}{2N_c} \int_{x,y,x',w'} \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} \left[\int_{\text{UV}} \frac{d^2 z}{(z - x)^2} \right] \times (\text{colour} \times \rho\rho).$$

That is $\mathcal{P}_{gg}$ times the product of the two LO Weizsäcker–Williams kernels — the DGLAP form — with no $d_\perp$ left anywhere in it. Note how differently it arises from Row IV: there C_{UV} came out of the p^+ bracket, here it is the *transverse* residue.

6. Two more logarithms, and which of them is real

$|z| \rightarrow \infty$: **a genuine infrared log**

This one is easy to miss. Three terms grow linearly in $|z|$ — T3's $\frac{(y-x) \cdot (y-z)}{2(y-x)^2}$, T4's $\frac{(y-x)^m (y-z)^i}{2(y-x)^2}$ and T6's $\frac{(y-x)^i (y-z)^m}{2(y-x)^2}$, all of which carry $(y - z)$ over a *fixed* $(y - x)^2$. Against a kernel $\sim 1/|z|$ and a denominator $\sim q z^2$ that leaves $1/z^2$, so $\int d^2 z \sim \int \frac{d|z|}{|z|}$. Residue (verified for $d_\perp = 2, 3, 4$ and two η , six cases, all matching to four digits):

$$Z^2 \langle \text{integrand} \rangle \longrightarrow \frac{V \cdot R}{2k^+ R^2} \left[\frac{1}{d_\perp} \frac{k^+ - p^+}{p^+} + \frac{k^+}{k^+ - p^+} \right].$$

It is *not* C_{UV} , and its $d_\perp$ does not cancel — a genuinely different divergence. This is the one that has to cancel against the $\tilde{\mathfrak{B}}_2$ (virtual) term of eq. (1), where $U(z) \rightarrow 1$ at large $|z|$: the same BK mechanism as in Row IV. **Which is why the colour collapse of §4 must not be made before this region is dealt with** — it throws away the $U(z)$ that does the cancelling.

$|y| \rightarrow \infty$: **not a divergence of the loop**

The angular average falls as $1/y^2$ (by exact quadrature: $y^2 \langle \cdot \rangle = 0.0862956$ at $|y| = 10^2, 10^3, \dots, 10^6$, constant to six digits), so $\int d^2 y$ would be log divergent too. But y is a **source** position — it sits under $\rho^f(y)$, inherited from $\mathfrak{B}_2$'s $\int_{\mathbf{x}} \rho^a(\mathbf{x})$ — so that tail is controlled by the target correlator $\langle \rho\rho \rangle$, not by a regulator. Same for x' under $\rho^{b'}(x')$. Not something to dimensionally regulate.

Regions that are safe

$y \rightarrow x$, $y \rightarrow z$ and $x' \rightarrow w'$ all give $u^2 \langle \cdot \rangle \rightarrow 0$ — integrable. The joint region $z, y \rightarrow x$ is log divergent by power counting (integrand $\sim r^{-3}$ against $d^2 y d^2 z \sim r^4 \frac{dr}{r}$), but it is the *same* logarithm: the $z \rightarrow x$ residue behaves as $\frac{V \cdot R}{R^2} \sim \frac{1}{|R|}$, which is integrable against $d^2 y = |R| d|R| d\phi$. So no $\log^2$ from the transverse side.

$p^+ \rightarrow 0$ and $p^+ \rightarrow k^+$

$C_{\text{UV}}(\eta)$ has $\frac{1}{\eta}$ and $\frac{1}{\bar{\eta}}$ poles at the two endpoints. These are **rapidity** divergences, not transverse ones — exactly what your $\int_{\Lambda}^{k^+ - \Lambda} dp^+$ is for. They will give the $\log \frac{k^+}{\Lambda}$ and the plus prescription, as in Row IV.

7. Fix list

1. **Keep** $U(z)$. Replace eq. (3)'s colour bracket by $[f^{adc} f^{ebf} U^{db}(x) U^{ce}(z) + N_c U^{af}(y)]$ and make $U(z) \rightarrow U(x)$ only inside the UV subtraction, where it is exact. Otherwise the large- $|z|$ cancellation is lost.
2. $\int_{w,w'} \rightarrow \int_{w'}$ in eq. (3) — the $\delta^{(2)}$ used the w integral.
3. $U^{b'a}(w')$ **versus** $U^{ab'}(w')$ in the first bracket: fix the index order (and drop the redundant δ^{ad} in the second colour term).
4. **Define** $\bar{\mathfrak{A}}, \bar{\mathfrak{B}}, \bar{\mathfrak{C}}$ in the note. The overall sign of eq. (2) rests entirely on them: the vertex product alone gives $-\frac{g^4}{8\pi^4} \frac{p^+ q}{k^{+3}}$, and the $+$ you wrote needs the first bracket to supply the second minus.
5. eq. (1) writes $\int_{x,z}$; say explicitly that y comes from $\mathfrak{B}_2$'s source integral and x' from $\mathfrak{A}^{(1)}$'s, which is how you legitimately get $\int_{x,y,z,x'}$ in eq. (2).

Checks: `src/rowB2_check.py` — §1 the six blocks in $d_\perp = 2, 3, 4, 5$; §2 the prefactor in sympy; §3 the colour identities with real adjoint $SU(3)$ lines; §4 the p^+ weight transfer; §5 the UV residue; §6 the region scan, the large- $|z|$ residue and the large- $|y|$ quadrature.